

01 **SNAPDRAGON 888**

If you are looking at what smartphones in the year ahead can look like, a good place to start is to keep an eye on the offerings by the component manufacturers. This is because only a handful of companies provide the components for making all the smartphones. For example, the sensors are from Samsung or Sony, and the Glass is always from Corning with the rare exception using Asahi. So, look at the offerings of the component manufacturers, and you get a pretty good idea of what is in store for the year ahead. When it comes to smartphones, the SoC that will set the benchmark for all flagships in 2021 will be the Snapdragon 888, manufactured with the latest 5nm process. In Dec 2020, Qualcomm announced the Snapdragon 888 at the Snapdragon Tech Summit Digital 2020, which was of course a virtual event. The Snapdragon 888 has a newly designed AI engine aimed at making everything from AI photography to personal assistants better. There is a low-powered AI processor to enable less demanding use cases such as screen awake as well as lift, activity and audio detection. The platform is powerful enough to record 4K videos from three sensors at the same time, and can fuse the information from three sensors to produce one superior image. The platform features the new Qualcomm Spectra 580 Image Signalling Processor, which seems to have been built just to allow smartphone manufacturers to make camera sensors too small to be allowed, then extract the light information from these tiny sensors to produce awesome images anyway. The Spectra 580 ISP can process images at a breathtaking 2.7 gigapixels per second, so there is no appreciable waiting time for all the magic.

Alex Katouzian, Senior Vice-President and General Manager of Mobile, Compute and Infrastructure at Qualcomm Technologies said, "Qualcomm's history of breakthrough technology inventions has paved the way for our continued leadership in the premium-tier mobile segment. The new flagship Snapdragon 888

Mobile Platform will enable OEMs to further differentiate their devices and allows users to experience the latest cutting-edge mobile technologies."

Lei Jun, Founder, chairman and chief executive officer of Xiaomi said, "The Snapdragon 888 is the most powerful mobile platform from Qualcomm Technologies ever. In addition to the industry leading 5G connectivity, it has brought several groundbreaking breakthroughs and innovations in AI, gaming, and camera. I'm glad that our new flagship smartphone 'Mi 11' will be the one of the first devices with Snapdragon 888. This is another cutting-edge product from us and will be loaded with various hardcore technologies."

Mitsuya Kishida, president of Sony Mobile Communications Inc. said "Xperia smartphones are feature-packed with Sony's advanced imaging and entertainment technologies, and it's essential for



The Snapdragon 888

Credit: Qualcomm

our products to be powered by the latest premium Snapdragon mobile platform to offer the best-in-class experiences to our fans. One of the entertainment experiences which we are very passionate about is mobile gaming, and we are overwhelmed by the incredible positive response we've received from fans around the world for Xperia 1 II and Xperia 5 II."

Devices featuring the Snapdragon 888 can be expected to start rolling out in the first quarter of 2021.

02 **INCREASED 5G ADOPTION**

A forecast from the *International Data Corporation* indicates that sales of 5G smartphones is all set to explode over the next few years, with 2021

showing the highest growth rate. From a depressing year, smartphones sales picked up in the third quarter of 2020, and are only expected to increase in 2021. Ryan Reith, Program Vice-President with IDC's Worldwide Mobile Device Trackers says, "Despite concerns around weakness in 5G demand, smartphone volumes exceeded the forecast in Q3-20 and supply-side momentum headed into the holiday quarter and 2021 remains strong. We've come to the conclusion that despite on-going lockdowns and economic concerns, consumers in many markets around the world have shifted their normal spending from things like travel, dining out, and general leisure to things like consumer electronics. Smartphones happen to be a benefactor of this transition."

We have manufacturers trying to reduce the cost of 5G hardware, and bring the prices of 5G smartphones at par with 4G smartphones. These efforts expected to significantly boost the sales of 5G smartphones.

Sangeetika Srivastava, senior analyst with IDC's Worldwide Mobile Device Trackers explains, "Competitive pricing will play an integral role in shaping 5G development. The COVID-19 crisis has influenced consumer behavior by tilting it toward more budget-friendly devices and narrowing the spend for essentials only. Aggressive promotions and more affordable 5G devices from major smartphone vendors are expected to partially offset the impact in the near term."

At the same time we have cutting edge technologies pushing the capabilities of 5G devices, such as the Snapdragon X60 5G Modem on the new Snapdragon 888 chip which can deliver speeds of 7.5 Gbps. Naoki Tani, Executive Vice-President and Chief Technology Officer, NTT DCOMO, INC says, "NTT DCOMO has been a leading mobile innovator in Japan for nearly 30 years and this leadership has continued into the 5G era, as the first operator in Japan to launch commercial 5G service in March this year (2020). 5G enables a new era of mobile experiences